

**UNIVERSITY OF SKILLS TRAINING AND ENTREPRENEURIAL DEVELOPMENT
(USTED)**

PROJECT PROPOSAL

PROJECT TITLE:

**Design and Development of a Web-Based Lost and Found Item Management System for
University Campuses**

Submitted by:

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1. INTRODUCTION

Losing personal belongings on university campuses is a common challenge faced by students, staff, and visitors. In many institutions, there is no centralized system for reporting lost items or notifying the rightful owners when such items are found. This often results in the permanent loss of valuable items and creates inconvenience for members of the university community.

This project seeks to develop a web-based Lost and Found Item Management System that will provide a secure and efficient platform for reporting, searching, and managing lost and found items within the university environment.

2. PROBLEM STATEMENT

Currently, the process of recovering lost items on campus is largely informal and ineffective. Information regarding found items is often communicated verbally or through social media platforms, making it difficult for rightful owners to retrieve their belongings. The absence of a centralized and well-organized system leads to poor record management, limited accessibility to information, and a low recovery rate of lost items.

3. AIM OF THE PROJECT

The aim of this project is to design and develop a web-based Lost and Found Item Management System that enables users to report lost or found items and facilitates the efficient recovery of misplaced belongings on campus.

4. OBJECTIVES OF THE PROJECT

The specific objectives of this project are to:

1. Develop a user-friendly platform for reporting lost items.
2. Enable users to upload details and images of found items.
3. Allow users to search for lost or found items using relevant filters.
4. Implement a secure authentication system for users and administrators.
5. Provide an administrative dashboard for monitoring and managing all reports.
6. Improve the efficiency and transparency of the lost-and-found process within the university community.

5. SCOPE OF THE PROJECT

The system will primarily serve students within the university community. It will consist of two main modules:

User Module

- User registration and login.
- Reporting of lost items.
- Reporting of found items.
- Uploading images of found items.
- Searching and viewing reported items.
- Viewing the status of submitted reports.

Administrator Module

- Secure administrator authentication.
- Monitoring all reported items.
- Verifying and managing reports.
- Updating the status of items (claimed, pending, resolved).
- Removing inappropriate or fraudulent reports.
- Generating basic reports and statistics.

6. SIGNIFICANCE OF THE STUDY

The proposed system will:

- Provide a centralized platform for lost and found item management.
- Increase the chances of recovering lost belongings.
- Reduce reliance on informal communication channels.
- Enhance accountability and transparency in handling found items.
- Improve the overall experience of students regarding lost property issues.

7. METHODOLOGY

The project will adopt the Agile Software Development methodology, allowing iterative development, testing, and continuous improvement throughout the project lifecycle.

The development process will include:

- Requirement gathering and analysis.
- System design and prototyping.
- User interface design.
- System development.
- Testing and debugging.
- Deployment and evaluation.

8. SYSTEM ARCHITECTURE AND TECHNOLOGIES

Frontend Development

- **Framework:** Next.js
- **User Interface Design:** Figma

Backend Development

- **Runtime Environment:** Node.js
- **Framework:** Express.js

Database

- **MongoDB**

Authentication and Security

- **JSON Web Tokens (JWT)**

Image Storage

- **Cloud nary**

Version Control

- **Git and GitHub**

9. EXPECTED OUTPUT

The completed system is expected to:

- Allow users to register and securely access the platform.
- Enable users to report lost and found items.
- Support image uploads for found items.
- Provide search functionality to help users locate their belongings.
- Equip administrators with tools to manage the platform effectively.

10. CONCLUSION

The proposed Lost and Found Item Management System seeks to address the challenges associated with recovering lost belongings within the university community. By leveraging modern web technologies, the system will provide a secure, efficient, and user-friendly solution that improves communication, accountability, and the likelihood of successfully reuniting owners with their lost items.

The successful implementation of this project will demonstrate the practical application of Information Technology in solving real-world problems within educational institutions.

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Supervisor's Approval

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